

$$\ln y = ax + by. \quad b = \frac{\ln y - ax}{y}.$$

$$0 = \left(\frac{\ln y - ax}{y} \right)' = \frac{\left(\frac{y'}{y} - a \right)y - (\ln y - ax)y'}{y^2} = \frac{y'(1 - \ln y + ax) - ay}{y^2};$$

$$a = \frac{y'(\ln y - 1)}{xy' - y}.$$

$$0 = \frac{\left(y''(\ln y - 1) + \frac{(y')^2}{y} \right)(xy' - y) - y'(\ln y - 1)(y' + xy'' - y')}{(xy' - y)^2}.$$

$$xy'y''(\ln y - 1) + \frac{x(y')^3}{y} - yy''(\ln y - 1) - (y')^2 - xy'y''(\ln y - 1) = 0.$$

$$y^2y''(\ln y - 1) = (y')^2(xy' - y).$$